Semester 5 · Mechanical / Tool & Die / Manufacturing · 4 Credits

Modern Production Processes

Professional handbook for jigs and fixtures, powder metallurgy, surface modification, non-conventional machining, NC/CNC programming, rapid prototyping, CIM, FMS and industrial robotics.

Course purpose

This subject shows how modern factories improve productivity using special tooling, advanced machining, CNC, rapid prototyping and computer integrated manufacturing.

Malayalam: Modern production-□ mass production accuracy, non-conventional machining, CNC control, robotics, FMS □□□□□□□□□□□ productivity □□□□□□□□□□□□□□.

Official details

Code: 5023A · **Title:** Modern Production Processes · **Category:** Program Elective / Core / Core · **Periods:** 60.

Prerequisites: Basic CAD Lab, Advanced CAD Lab, Machine Tools and Mechanical Workshop.

CO1

Jigs, fixtures, powder metallurgy and surface modification.

CO2

Non-conventional machining principles and applications.

CO3

NC/CNC machines and ISO part programming.

CO4

CIM, DNC, CAPP, FMS, AGV and robots.

Roadmap

CO1 · 12 Hours

Jigs, Fixtures, Powder Metallurgy and Surface Modification

Jigs and fixtures

Jigs locate, hold and guide the tool; fixtures locate and hold the work but normally do not guide the tool. They reduce setting time, improve repeatability and make mass production economical. Types include box jig, indexing jig, angle plate jig and channel jig. Fixtures are used for turning, drilling, milling and grinding.

Powder metallurgy

Powder metallurgy uses powder production, blending, compacting, sintering and finishing. It is suitable for self-lubricating bearings, filters, hard materials and complex small parts. Advantages are material saving and near-net shape; limitations include equipment cost and size restriction.

Surface modification

Surface methods improve wear, corrosion and fatigue resistance without changing the whole component. Methods include PVD, CVD, diffusion coating, metal spraying and organic coatings.

Expected questions

1. Differentiate jig and fixture.
2. Explain powder metallurgy steps.
3. Explain PVD, CVD and metal spraying with sketches.

CO2 · 17 Hours

Non-Conventional Machining Processes

Need and classification

Non-conventional machining is used for hard, brittle, heat-sensitive or complex materials where ordinary cutting is difficult. Processes may be mechanical, thermal, electrical, chemical or electrochemical.

USM, EDM and WEDM

USM removes material by abrasive slurry and ultrasonic vibration. EDM removes conductive material by spark erosion in dielectric fluid. Wire-cut EDM uses a moving wire electrode to cut accurate profiles.

AWJM, LBM and ECM

Abrasive water jet cuts using high velocity abrasive jet. Laser beam machining uses focused thermal energy. Electrochemical machining removes metal by anodic dissolution without tool wear.

Expected questions

1. Classify non-conventional machining.
2. Explain EDM equipment and dielectric function.
3. Compare EDM and ECM.
4. List applications, advantages and limitations of AWJM and LBM.

CO3 · 14 Hours

NC, CNC, Part Programming and Rapid Prototyping

NC and CNC

CNC machines use computer control to move axes, control spindle, change tools and execute programs. Classification includes motion type, control loop and axis. Machining centres combine automatic tool changer and tool magazine.

Part programming

ISO part program uses preparatory functions G and miscellaneous functions M. A good program has safe start, tool selection, spindle speed, feed, coordinate movement, machining blocks and safe return.

Example structure: G21 G90; tool call; spindle ON; rapid approach; feed cut; retract; M30.

RP and 3D printing

Rapid prototyping converts CAD model to physical model layer by layer. Steps include CAD model, STL conversion, slicing, layer deposition and post-processing.

Expected questions

1. Compare NC and CNC.
2. Explain machine axes and motion control.
3. Write a simple turning, drilling or milling part program.
4. Explain RP steps and applications.

CO4 · 15 Hours

CIM, FMS and Robotics

CIM and CAPP

CIM integrates design, process planning, production, inspection, storage and business data. DNC links multiple machines to a central controller. CAPP prepares process plans using computer support.

FMS

Flexible manufacturing system uses CNC machines, material handling, computer control and group technology. Layouts include single line, dual line, loop, ladder, carousel and robot-centered layouts. AGV supports automatic material transport.

Robotics

Robot anatomy includes manipulator, joints, links, end effector, drive, controller and sensors. Configurations include Cartesian, cylindrical, spherical, SCARA and articulated robots.

Expected questions

1. Explain CIM structure and benefits.
2. Explain FMS components and layouts.
3. Explain group technology and part family.
4. Explain robot anatomy and configurations.

Rule bank

Jig guides tool; fixture holds only. EDM needs conductive workpiece. ECM has no tool wear in ideal condition. CNC program must define units, coordinates, tool, spindle, feed and safe end.

Model paper

1. Explain jigs, fixtures and powder metallurgy.
2. Explain EDM, WEDM, AWJM, LBM and ECM.
3. Write an ISO CNC part program for a simple operation.
4. Explain CIM, FMS layout and robot anatomy.

Answer guide

For process answers write principle, equipment, working, material suitability, advantages, limitations and applications. For CNC programming, keep sequence safe and include comments in exam answer. For CIM/FMS answers draw block or layout diagrams.

References: HMT Production Technology, R.K. Jain, Pabla and Adithan, Jagadeesha T, Ian Gibson, Mikell Groover and NPTEL resources.